

BACKEND SCHEMA DOCUMENT

Project Name

WorkForce360

Database Overview

Database Engine

PostgreSQL

ORM

Prisma ORM

Schema Design Philosophy

- Fully Normalized Database
 - UUID-based Primary Keys
 - Soft Delete Strategy
 - Audit-Friendly Structure
 - Payroll Freeze Support
 - Invoice Freeze Support
 - Historical Data Preservation
-

GLOBAL CONVENTIONS

Primary Key

Every table must contain:

`id UUID PRIMARY KEY`

Audit Fields

Every major table must contain:

```
created_at  TIMESTAMP
updated_at  TIMESTAMP
created_by  UUID
updated_by  UUID
```

Soft Delete

Use:

```
is_active  BOOLEAN
```

instead of deleting records.

AUTHENTICATION MODULE

users

Purpose:

Stores Admin and Worker login accounts.

Columns

Column	Type
id	UUID
employee_id	VARCHAR(20)
password_hash	TEXT
role	ENUM
last_login	TIMESTAMP
is_active	BOOLEAN
created_at	TIMESTAMP

Column	Type
updated_at	TIMESTAMP

Role Values

ADMIN
WORKER

Indexes

employee_id UNIQUE

user_sessions

Purpose:

Track active JWT sessions.

Columns

Column	Type
id	UUID
user_id	UUID
token	TEXT
expires_at	TIMESTAMP
created_at	TIMESTAMP

Relationships

user_id → users.id

MASTER TABLES

designations

Purpose:

Manage employee designations.

Columns

Column	Type
id	UUID
name	VARCHAR(100)
description	TEXT
is_active	BOOLEAN

expense_categories

Purpose:

Manage expense categories.

Columns

Column	Type
id	UUID
name	VARCHAR(100)
description	TEXT
is_active	BOOLEAN

EMPLOYEE MODULE

employees

Purpose:

Main employee profile.

Columns

Column	Type
id	UUID

Column	Type
employee_code	VARCHAR(20)
full_name	VARCHAR(255)
age	INTEGER
dob	DATE
nationality	VARCHAR(100)
gender	VARCHAR(20)
religion	VARCHAR(100)
phone_number	VARCHAR(20)
email	VARCHAR(255)
present_address	TEXT
permanent_address	TEXT
aadhar_number	VARCHAR(20)
pf_number	VARCHAR(50)
esic_number	VARCHAR(50)
joining_date	DATE
designation_id	UUID
status	ENUM
current_contract_id	UUID
is_active	BOOLEAN

Status Values

ACTIVE
INACTIVE
RESIGNED
TERMINATED

Relationships

designation_id → designations.id

current_contract_id → contracts.id

Indexes

employee_code UNIQUE

phone_number

aadhar_number

pf_number

esic_number

employee_bank_details

Columns

Column	Type
id	UUID
employee_id	UUID
bank_name	VARCHAR(255)
account_holder_name	VARCHAR(255)
account_number	VARCHAR(100)
ifsc_code	VARCHAR(20)
branch_name	VARCHAR(255)

Relationships

employee_id → employees.id

employee_family_members

Columns

Column	Type
id	UUID
employee_id	UUID
relationship	ENUM

Column	Type
full_name	VARCHAR(255)
occupation	VARCHAR(255)
dob	DATE
contact_number	VARCHAR(20)

Relationship Values

FATHER
MOTHER
SPOUSE
CHILD

employee_education

Columns

Column	Type
id	UUID
employee_id	UUID
qualification	VARCHAR(255)
institution	VARCHAR(255)
specialization	VARCHAR(255)
completion_year	INTEGER
score	DECIMAL(5,2)
score_type	ENUM

Score Type

PERCENTAGE
GPA

employee_employment_history

Columns

Column	Type
id	UUID
employee_id	UUID
employer_name	VARCHAR(255)
designation	VARCHAR(255)
experience_years	DECIMAL(5,2)

employee_documents

Columns

Column	Type
id	UUID
employee_id	UUID
document_type	ENUM
cloudinary_url	TEXT
uploaded_at	TIMESTAMP

Document Types

PHOTO
AADHAAR
PAN
PASSBOOK
JOINING_FORM
PF
ESIC

SALARY MODULE

salary_structures

Purpose:

Current salary configuration.

Columns

Column	Type
id	UUID
employee_id	UUID
fixed_gross	DECIMAL(12,2)
tea_allowance	DECIMAL(12,2)
basic_percentage	DECIMAL(5,2)
hra_percentage	DECIMAL(5,2)
conveyance_percentage	DECIMAL(5,2)
fixed_basic	DECIMAL(12,2)
fixed_hra	DECIMAL(12,2)
fixed_conveyance	DECIMAL(12,2)
effective_from	DATE

salary_revisions

Purpose:

Maintain salary history.

Columns

Column	Type
id	UUID
employee_id	UUID

Column	Type
salary_structure_id	UUID
revision_date	DATE
remarks	TEXT

CLIENT MODULE

clients

Columns

Column	Type
id	UUID
company_name	VARCHAR(255)
gst_number	VARCHAR(50)
pan_number	VARCHAR(50)
address	TEXT
contact_person	VARCHAR(255)
phone_number	VARCHAR(20)
email	VARCHAR(255)
is_active	BOOLEAN

CONTRACT MODULE

contracts

Columns

Column	Type
id	UUID
client_id	UUID

Column	Type
contract_name	VARCHAR(255)
start_date	DATE
end_date	DATE
service_charge_percentage	DECIMAL(5,2)
cgst_percentage	DECIMAL(5,2)
sgst_percentage	DECIMAL(5,2)
terms_conditions	TEXT
status	ENUM

Status

DRAFT
ACTIVE
EXPIRED
TERMINATED

Relationships

client_id → clients.id

ATTENDANCE MODULE

attendance

Purpose:

Daily attendance records.

Columns

Column	Type
id	UUID
employee_id	UUID
attendance_date	DATE

Column	Type
attendance_status	ENUM
overtime_hours	DECIMAL(5,2)
remarks	TEXT

Status

```
PRESENT
HALF_DAY
ABSENT
```

Indexes

employee_id

attendance_date

employee_id + attendance_date UNIQUE

PAYROLL MODULE

payroll_batches

Purpose:

Monthly payroll runs.

Columns

Column	Type
id	UUID
payroll_month	INTEGER
payroll_year	INTEGER
total_payroll_cost	DECIMAL(14,2)
total_contributions	DECIMAL(14,2)
payroll_generated_at	TIMESTAMP

Column	Type
is_frozen	BOOLEAN

payroll_items

Purpose:

Employee payroll snapshots.

Columns

Column	Type
id	UUID
payroll_batch_id	UUID
employee_id	UUID
total_days	INTEGER
payable_days	DECIMAL(5,2)
absent_days	DECIMAL(5,2)
overtime_hours	DECIMAL(8,2)
earned_basic	DECIMAL(12,2)
earned_hra	DECIMAL(12,2)
earned_conveyance	DECIMAL(12,2)
earned_tea_allowance	DECIMAL(12,2)
overtime_wages	DECIMAL(12,2)
special_allowance	DECIMAL(12,2)
gross_pay	DECIMAL(12,2)
pf	DECIMAL(12,2)
esic	DECIMAL(12,2)
professional_tax	DECIMAL(12,2)
net_salary	DECIMAL(12,2)
employer_pf	DECIMAL(12,2)
employer_pf_admin	DECIMAL(12,2)

Column	Type
employer_esic	DECIMAL(12,2)
total_ctc	DECIMAL(12,2)

Relationships

payroll_batch_id → payroll_batches.id

employee_id → employees.id

PAYSLIP MODULE

payslips

Columns

Column	Type
id	UUID
payroll_item_id	UUID
pdf_url	TEXT
generated_at	TIMESTAMP

INVOICE MODULE

invoice_batches

Purpose:

Monthly invoices.

Columns

Column	Type
id	UUID
contract_id	UUID

Column	Type
payroll_batch_id	UUID
invoice_number	VARCHAR(100)
invoice_date	DATE
total_invoice_amount	DECIMAL(14,2)
is_frozen	BOOLEAN

invoice_items

Columns

Column	Type
id	UUID
invoice_batch_id	UUID
payroll_cost	DECIMAL(14,2)
employer_pf	DECIMAL(14,2)
employer_pf_admin	DECIMAL(14,2)
employer_esic	DECIMAL(14,2)
service_charge	DECIMAL(14,2)
cgst_amount	DECIMAL(14,2)
sgst_amount	DECIMAL(14,2)
total_amount	DECIMAL(14,2)

EXPENSE MODULE

expenses

Columns

Column	Type
id	UUID

Column	Type
category_id	UUID
expense_name	VARCHAR(255)
amount	DECIMAL(12,2)
expense_date	DATE
description	TEXT

Relationships

category_id → expense_categories.id

SETTINGS MODULE

settings

Columns

Column	Type
id	UUID
company_name	VARCHAR(255)
company_logo_url	TEXT
address	TEXT
phone_number	VARCHAR(20)
email	VARCHAR(255)
gst_number	VARCHAR(50)
pan_number	VARCHAR(50)
default_pf_percentage	DECIMAL(5,2)
default_pf_admin_percentage	DECIMAL(5,2)
default_esic_percentage	DECIMAL(5,2)

AUDIT MODULE

audit_logs

Purpose:

Track critical system activities.

Columns

Column	Type
id	UUID
user_id	UUID
action	VARCHAR(255)
entity_type	VARCHAR(100)
entity_id	UUID
description	TEXT
created_at	TIMESTAMP

Examples:

- Payroll Generated
- Invoice Generated
- Salary Updated
- Password Reset
- Employee Deactivated

PERMISSIONS

ADMIN

Full CRUD Access

All Modules

WORKER

Read Own Profile

Read Own Attendance

Read Own Payroll

Download Own Payslips

Change Own Password

No Edit Access

DATA OWNERSHIP RULES

Employee Data

Owned By: Admin

Visible To: Admin + Respective Employee

Payroll Data

Owned By: Admin

Visible To: Admin + Respective Employee

Invoices

Owned By: Admin

Visible To: Admin Only

Expenses

Owned By: Admin

Visible To: Admin Only

FREEZE RULES

Payroll

Once Generated:

- Cannot Regenerate
 - Cannot Modify
 - Cannot Delete
-

Invoice

Once Generated:

- Cannot Regenerate
 - Cannot Modify
 - Cannot Delete
-

INDEXING STRATEGY

Create Indexes On:

employees.employee_code

employees.phone_number

attendance.employee_id

attendance.attendance_date

payroll_batches.payroll_month

payroll_batches.payroll_year

invoice_batches.invoice_number

clients.company_name

contracts.status

expenses.expense_date

audit_logs.created_at

PERFORMANCE STRATEGY

Target Support:

- 500+ Employees
- 100+ Contracts
- 5+ Years Historical Payroll Data

Optimizations:

- Indexed Search
- Paginated Queries
- Lazy Loading Documents
- Aggregated Dashboard Queries
- Frozen Payroll Snapshots
- Frozen Invoice Snapshots